Interactive Activation Model(IAM) as Bayesian Inference

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Abstract

The Interactive Activation Model (IAM), introduced by (McClelland & Rumelhart, 1985), marked a significant paradigm shift in cognitive computation, transitioning from static to dynamic interaction models within neural networks. This model, originally applied to encode relational data from the 1960s musical "West Side Story," utilizes a network where node activations influence one another and evolve over time to achieve an equilibrium state. Such dynamics allow the model to exhibit cognitive behaviors akin to human pattern recognition, semantic priming, and associative memory retrieval. Despite its innovative approach, the model has faced criticism, particularly from Pearl's later development of a unified Bayesian framework, which questioned the validity of inference mechanisms in connectionist models due to issues like incomplete bi-directional information propagation. Continuing from these critiques, our research reevaluates the IAM through the lens of Bayesian inference by reconstructing it within a Bayesian Network, employing a naive Bayesian classifier. This study not only revisits the IAM's capabilities such as content addressability, graceful degradation, and spontaneous generalization but also aims to bridge the gap between traditional connectionist approaches and contemporary Bayesian methods. Through this integration, we explore the foundational relationship between connectionist models and Bayesian reasoning, providing new insights into their interplay and implications for cognitive science and artificial intelligence.

Keywords: Cognitive Modeling; Iteractive Activation Model; Bayesian Network; Naive Bayes; Probabilistic Modeling

Introduction

McClelland and Rumelhart proposed the Interactive Activation Model (IAM) (McClelland & Rumelhart, 1985) that represents a significant departure from earlier static models of cognitive computation. At its core, the IAM proposes a dynamic and interactive system where in the activation of one node within a network influences the activation levels of neighboring nodes. McClelland and Rumelhart, in their paper, proposed a model encoding the knowledge in the 1960s musical "West Side Story," where two gangs struggled for the control of Manhattan. This model maps the instance of gangsters to their properties including their gang names, education, age, marriage status and occupations. The connection would get excited when the corresponding property label matches the instance and would be inhibitory when they do not match. The activation of these units would evolve with respect to time and would reach equilibrium at some time. In this computational model, some interesting patterns have been discovered, including pattern recognition, semantic priming, and associative memory retrieval, which are found to be similar to human cognitive behaviors.

However, later Pearl (Pearl, 1982) proposed a unified Bayesian framework that transformed previous connectionist processing models into a Bayesian inference one. In comparison to the Bayesian model, several concerns have been raised about the validity of inference in connectionist processing models (McClelland, 2013), especially about the missing of information in the bi-directional propagation of information. For instance, (Massaro, 1989) pointed out that Interactive Activation is not necessarily in explaining speech recognition and is in fact incorrect in the TRACE model, which is a connectionist model in speech recognition.

Reflecting on these developments, the work of Tenenbaum et al. (Tenenbaum, Kemp, Griffiths, & Goodman, 2011) on the Bayesian frameworks further illuminates the potential of these models to grasp abstract knowledge from sparse data. Their review underscores a significant shift towards understanding the generative nature of cognitive processes, aligning well with the principles underlying both the IAM and newer Bayesian models.

As a continuing work, we aim to evaluate the IAM presented in McClelland and Rumelhart's paper within a Bayesian Inference network framework. We first build the entire model in a Bayesian Network fashion by using a "naive Bayesian classifier". Furthermore, we evaluate several interesting phenomena observed in IAM including content addressability, graceful degradation, and spontaneous generalization. Finally, we discuss the relationship between IAM and Bayesian modeling.

Methods

Dataset

In our experiment, we use the Jets and the Sharks dataset as used by McClelland. (McClelland & Rumelhart, 1985) The datset stems from the 1960s musical "West Side Story," where two gangs the "Jets" and "Sharks" struggle for neighborhood control in Manhattan. The dataset contains personal information for 27 gang members. For each gang member, their name, gang name, age, education status, marriage status and occupation is included in the dataset, where the detail of the dataset is included in Figure 1. All of the columns of the dataset are presented in categories. Later we will train our

The Jets and The Sharks

Name	Gang	Age	Edu	Mar	Occupation
Art	Jets	40's	J.H.	Sing.	Pusher
Al	Jets	30's	J.H.	Mar.	Burglar
Sam	Jets	20's	COL.	Sing.	Bookie
Clyde	Jets	40's	J.H.	Sing.	Bookie
Mike	Jets	30's	J.H.	Sing.	Bookie
Jim	Jets	20's	J.H.	Div.	Burglar
Greg	Jets	20's	H.S.	Mar.	Pusher
John	Jets	20's	J.H.	Mar.	Burglar
Doug	Jets	30's	H.S.	Sing.	Bookie
Lance	Jets	20's	J.H.	Mar.	Burglar
George	Jets	20's	J.H.	Div.	Burglar
Pete	Jets	20's	H.S.	Sing.	Bookie
Fred	Jets	20's	H.S.	Sing.	Pusher
Gene	Jets	20's	COL.	Sing.	Pusher
Ralph	Jets	30's	J.H.	Sing.	Pusher
Phil	Sharks	30's	COL.	Mar.	Pusher
Ike	Sharks	30's	J.H.	Sing.	Bookie
Nick	Sharks	30's	H.S.	Sing.	Pusher
Don	Sharks	30's	COL.	Mar.	Burglar
Ned	Sharks	30's	COL.	Mar.	Bookie
Karl	Sharks	40's	H.S.	Mar.	Bookie
Ken	Sharks	20's	H.S.	Sing.	Burglar
Earl	Sharks	40's	H.S.	Mar.	Burglar
Rick	Sharks	30's	H.S.	Div.	Burglar
Ol	Sharks	30's	COL.	Mar.	Pusher
Neal	Sharks	30's	H.S.	Sing.	Bookie
Dave	Sharks	30's	H.S.	Div.	Pusher

Figure 1: Information of a number of individuals belonging to two gangs, the Jets and the Sharks.

model in a supervised fashion on a classification task. For example, given the inference task $P(\text{Instance} \mid \text{Gang}, \text{Age}, \text{Edu})$, the model will be trained to predict the correct instance given the gang type, age and education information. After training, the parameters of the model will used to calculate the likelihood of attributes for each instance.

Furthermore, we preprocess each columns by transforming categorical string literals into numerical values for compatibility with the `Scipy` library. After inference, the numerical values are converted back to string for visualization.

Interactive Activation Model

The Interactive Activation Model (McClelland & Rumelhart, 1985) simulates the excitatory and inhibitory connections between neurons for information processing. During processing, one or more neurons are set as probes and are allowed to be active, firing signal to other neurons. The evolution of each neuron could be thought as a Recurrent Neural Network (Amari, 1972), where each neuron evolves on the dimension of time. At convergence the excitatory signals and the inhibitory signals tends to reach an equilibrium, indicating the model's decision given the probe's activation as input.

When updating the activation of each neuron, it first calculates the net signals coming from all connected signals. While at the same time, the neuron receiving the information fires

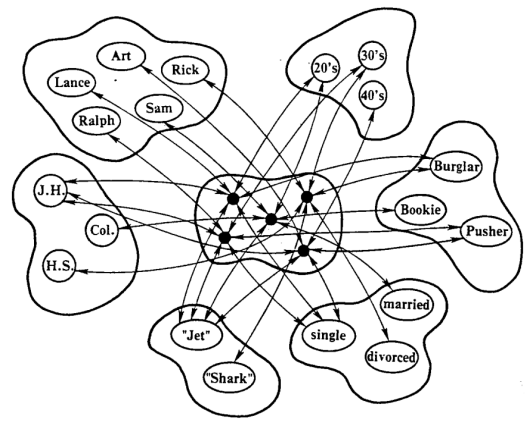

Figure 2: The connections between part of the individuals of gangsters to their properties. The arrows indicate the excitatory connections from the gangster instances to their properties, while the inhibitory connections are not shown.

signal back using the same equation where

$$y^{(t+1)} = g\left(y^{(t)}, \sum_i x_i^{(t)} w_i + b\right)$$

After summing up all the information from the neighborhood neurons, **net** incoming signals are sent into an activation function for update where

$$g\left(u^{(t)}, \text{net}\right) = \begin{cases} u^{(t)} + \left(\max - u^{(t)}\right) \text{net} - \text{decay}\left(u^{(t)} - rest\right) & \text{net} > 0 \\ u^{(t)} + \left(u^{(t)} - \min\right) \text{net} - \text{decay}\left(u^{(t)} - \text{rest}\right) & \text{net} \leq 0 \end{cases}$$

Here *decay*, *rest*, *max*, *min* are hyperparameters to be tuned.

In evaluation, multiple combinations of inputs are selected to observe the IAC model's output. It is interesting in that the model has several output patterns that is similar to human recognition patterns. Several examples include retrieval by name, retrieval by partial information, graceful degradation and spontaneous generalization.

The Bayes' Formula

Bayes' theorem, named after the Reverend Thomas Bayes, is a fundamental principle in probability theory that plays a pivotal role in cognitive psychology. It provides a mathematical framework for updating beliefs or hypotheses in light of new evidence, making it a cornerstone of reasoning under uncertainty. The Bayes' Formula follows the definition of conditional probability and is defined as:

$$p(h_i \mid e) = \frac{p(h_i)p(e \mid h_i)}{p(h_1)p(e \mid h_1) + p(h_2)p(e \mid h_2)}$$

where $p(h_i)$ is the prior probability, $p(e \mid h_i)$ is the probability that an event is true given hypothesis i, and $p(h_i \mid e)$ is the posterior. With these elements, posterior can be inferred

by having a prior and the conditional probability of an event given a hypothesis.

The application of Bayes formula is widespread in computational cognitive modelling. In perception modelling, individuals integrate sensory information with prior knowledge to form perceptual interpretations. Bayes' theorem provides a formal framework for modeling perceptual inference, where prior beliefs about the environment are updated based on incoming sensory evidence. For example, in visual perception, prior expectations about object shape or identity influence the interpretation of ambiguous or noisy visual stimuli.

Naive Bayes Model

The Naive Bayes classifier is a probabilistic machine learning model based on Bayes' theorem, which is used for classification tasks. It assumes that the features are conditionally independent given the class label, hence the term "naive." Despite this simplifying assumption, Naive Bayes often performs well in practice, especially for text classification and other high-dimensional datasets, as well as the tabular data case in our dataset.

Model Representation Let $X = (x_1, x_2, ..., x_n)$ denote a set of n features describing an instance, and let Y represent the class label. The Naive Bayes classifier calculates the posterior probability $P(Y|X)$ of a class label given the observed features using Bayes' theorem:

Assumption of Feature Independence The Naive Bayes classifier makes the naive assumption that the features $x_1, x_2, ..., x_n$ are conditionally independent given the class label Y. Mathematically, this is expressed as:

$$P(X|Y) = P(x_1|Y) \times P(x_2|Y) \times ... \times P(x_n|Y)$$

This assumption greatly simplifies the model and makes it computationally tractable, but it may not hold true in practice for all datasets.

Model Training and Evaluation To train the Naive Bayes classifier, we estimate the prior probabilities $P(instance)$ and the likelihoods $P(Attributes|instance)$ from the training data. During prediction, we calculate the posterior probability $P(instance|attributes)$ for each class *instance* using Bayes' theorem and select the class with the highest probability as the predicted class label.

Latent factor estimation The overall model is trained using the full dataset, where we predict $p(instance\,|\,attributes_{1...N})$. However, if the query is given as a subset of the attributes, we are interested in how to estimate $p(instance\,|\,attributes_{1...N-k})$ if only $N-k$ features are present. In our Naive Bayes model, we try to estimate $p(instance\,|\,attributes_{1...N-k})$ by modelling on the the subset of these attributes, then use the query to calculate the instance probability. Then our final activation can be estimatd as,

$$p(attributes\,|\,query) = p(instance\,|\,query)\,p(attributes\,|\,instance)$$

where the first term on the right hand side is modelled by the subset of attributes contained in the query and the second term is modelled by the entire dataset.

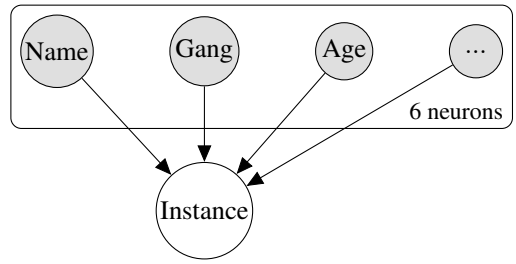

Figure 3: The Naive Bayes model approximating the IAM model. The instance node takes on values of all possible people in the dataset, while the neurons within the plate are the probability of attributes given the instance node. During inference, the model is trying to predict the value of the instance node given its attributes.

Results

Evaluation on Content Addressability, Graceful Degradation and Spontaneous Generalization

This probabilistic graphical model is evaluated based on content addressability, graceful degradation and spontaneous generalization. Content addressability is the ability to retrieve a name through partial information, graceful degradation is the ability to retrieve name through imperfect information and spontaneous generalization is the ability to retrieve common attributes for a specific group.

Content Addressability. We first test the model on its ability for retrieving name with partial information. Specifically, we want to get the name 'Ken' by giving the model input of 'Shark' and 'in 20s'. The table below shows each name and their corresponding latent probability.

Table 1: Latent Probabilities for names with properties 'Shark' and 'In 20s'

Name	Probability
Al	0.027
Art	0.027
Clyde	0.027
Dave	0.027
Don	0.027
Doug	0.027
Earl	0.027
Fred	0.027
Gene	0.027
George	0.027
Greg	0.027

continued on next column

continued from previous column

Name	Probability
Ike	0.027
Jim	0.027
John	0.027
Karl	0.027
Ken	**0.292**
Lance	0.027
Mike	0.027
Neal	0.027
Ned	0.027
Nick	0.027
OI	0.027
Pete	0.027
Phil	0.027
Ralph	0.027
Rick	0.027
Sam	0.027

As illustrated in Table 1, the majority of names are assigned a probability of 0.027, whereas Ken exhibits the highest probability at 0.292. This discrepancy suggests that the lower probability names are less likely given the specific input characteristics, notably being members of the "Sharks" and in their 20s. A review of the training dataset confirms that Ken is uniquely associated with these attributes, thereby corroborating the model's effectiveness in retrieving name by partial information.

Graceful Degradation. To evaluate on Graceful Degradation, we want to provide the model with imperfect information and examine its ability to retrieve the desired name. An example case is we want to retrieve the name "Ike" but give the model the below information:

- Sharks in their 30's
- Junior high education
- Bookie
- Divorced

However, in the dataset, the actual marriage status for Ike is "Single". Table 2 shows that Ike has the highest probability among all the other names, which indicate the model's ability to retrieve name even with imperfect information.

Table 2: Latent Probabilities for names with properties 'Shark' and 'In 30s'

Name	Probability
Al	0.027
Art	0.027
Clyde	0.027

continued on next column

continued from previous column

Name	Probability
Dave	0.027
Don	0.027
Doug	0.027
Earl	0.027
Fred	0.027
Gene	0.027
George	0.027
Greg	0.027
Ike	**0.296**
Jim	0.027
John	0.027
Karl	0.027
Ken	0.027
Lance	0.027
Mike	0.027
Neal	0.027
Ned	0.027
Nick	0.027
OI	0.027
Pete	0.027
Phil	0.027
Ralph	0.027
Rick	0.027
Sam	0.027

As demonstrated in Table 2, the name Ike has the highest probability of 0.296. This indicates that, based on the given properties, the model predicts Ike as the name most likely to be associated with the input. When compared with the training dataset, the model accurately predicts the name Ike, even though not all the provided properties match those associated with Ike. This effectively validates our model's performance with regard to graceful degradation.

Spontaneous Generalization. Now we consider the model's ability to retrieve appropriate generalizations over sets of individuals—that is, its ability to answer questions like "What are Jets like?"

Table 3: Spontaneous Generalization Results

Name	Probability
Al	0.045
Art	0.045
Clyde	0.045
Dave	0.027
Don	0.027
Doug	0.045
Earl	0.027
Fred	0.045
Gene	0.045

continued on next column

continued from previous column

Name	Probability
George	0.045
Greg	0.045
Ike	0.027
Jim	0.045
John	0.045
Karl	0.027
Ken	0.027
Lance	0.045
Mike	0.045
Neal	0.027
Ned	0.027
Nick	0.027
OI	0.027
Pete	0.045
Phil	0.027
Ralph	0.045
Rick	0.027
Sam	0.045

Group	Probability
Jets	**0.916**
cell5	0.084

Age	Probability
20's	**0.538**
30's	0.282
40's	0.179

Occupation	Probability
Bookie	0.333
Burglar	0.333
Pusher	0.333

Edu	Probability
COL.	0.179
H.S.	0.282
J.H.	**0.538**

Marital Status	Probability
Div.	0.167
Mar.	0.214
Sing.	**0.499**

Referring to Table 3, the probabilities associated with some names are 0.045, while others are 0.027. This differentiation aligns with expectations from the training data, where names are associated with either the 'Sharks' or 'Jets' gang. When 'Jets' is specified as input to the model, it logically assigns higher probabilities to names identified with the 'Jets' gang. Furthermore, the model assigns the highest probability to the age group '20s', which correlates well with the training data that shows a majority of the 'Jets' members are in their '20s'. The uniform probability distribution across occupations can be attributed to an equal representation of each occupation in the training dataset. These results underscore our model's capability for Spontaneous Generalization, demonstrating its effectiveness in making generalizations based on the input attributes.

Discussion

McClelland (2013) (McClelland, 2013) explores the integration of connectionist models like the interactive activation model with probabilistic modeling to enhance the understanding and simulation of cognitive processes. In our study, we further validated that interactive activation models can benefit from incorporating probabilistic principles by effectively handling uncertainty and ambiguous inputs through a Bayesian framework. This integration transforms these models to represent knowledge as dynamic probability distributions rather than fixed patterns, allowing for more adaptive learning and interpretation. Probabilistic elements enable the models to deal with noisy data and improve learning by treating network weights and structures as probabilistic, thus updating belief systems.

The findings from our study underscore the potential of hybrid models to not only understand but also to predict cognitive behaviors in more realistic scenarios. For instance, the ability to model spontaneous generalization and graceful degradation in connectionist frameworks can directly inform the development of more robust artificial intelligence systems. These systems could potentially exhibit greater resilience in the face of incomplete or evolving data sets, which is crucial for applications in dynamic environments such as autonomous vehicles and real-time data analysis.

Furthermore, the application of Bayesian networks in cognitive modeling extends beyond typical psychological experiments and into the realm of social and economic behaviors. By simulating how individuals might respond under different conditions, policymakers and organizations can better design interventions that are sensitive to the underlying probabilistic nature of human decision-making.

Looking forward, it is essential to address the scalability of these models. As we integrate more complex datasets and strive for higher accuracy in simulations, the computational demands increase. Future research should explore efficient algorithms that can handle large-scale networks without compromising the speed or accuracy of the probabilistic computations.

Additionally, there is a need for a standardized framework to evaluate and compare different hybrid models. This would not only help in benchmarking but also in identifying the best practices for integrating connectionist and probabilistic models across various fields of application.

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Appendix

The notebook of our experiment.